

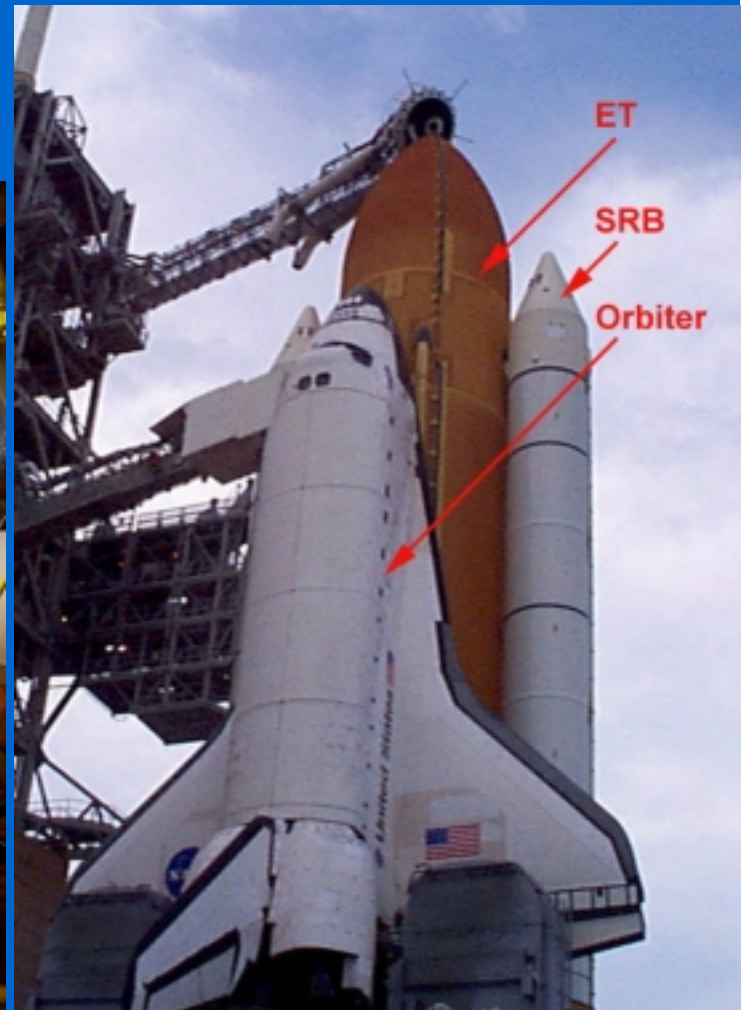
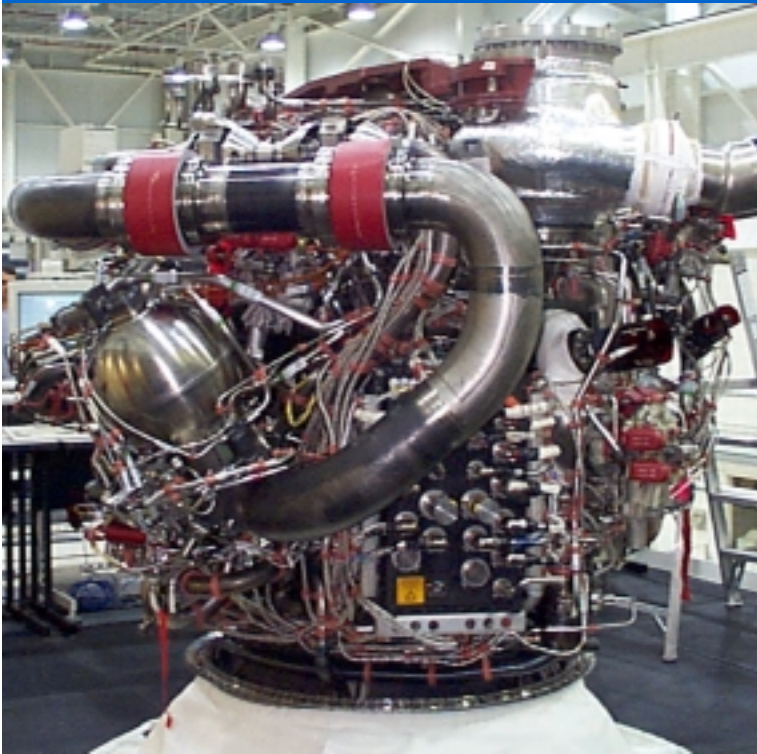
Quadrupole Ion Trap Mass Spectrometry for Space Shuttle Ground Support

A photograph of a space shuttle launching, with a large plume of smoke and fire at the base, set against a cloudy sky. The shuttle is ascending vertically, leaving a bright trail of fire and a large, billowing cloud of white smoke. The foreground shows a dark, flat landscape, possibly a field or marsh, with a body of water visible in the distance.

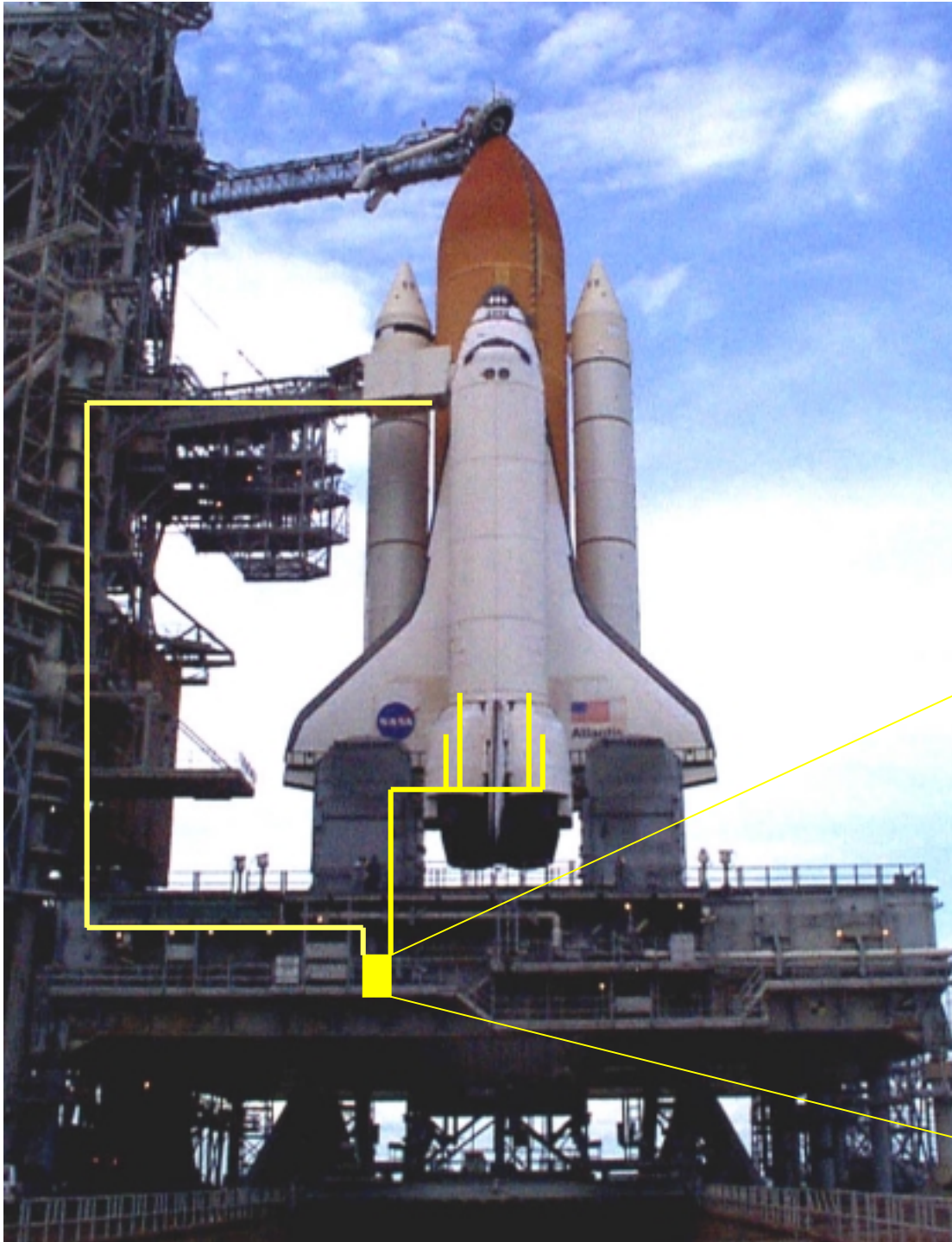
**Andrew K. Ottens, W. W. Harrison (University of Florida),
Timothy P. Griffin (Dynacs Engineering Inc.),
William R. Helms (NASA/ Kennedy Space Center)**

Cryogenic Fuel System

- 1,773,000 L LH_2
- 660,000 L LO_2

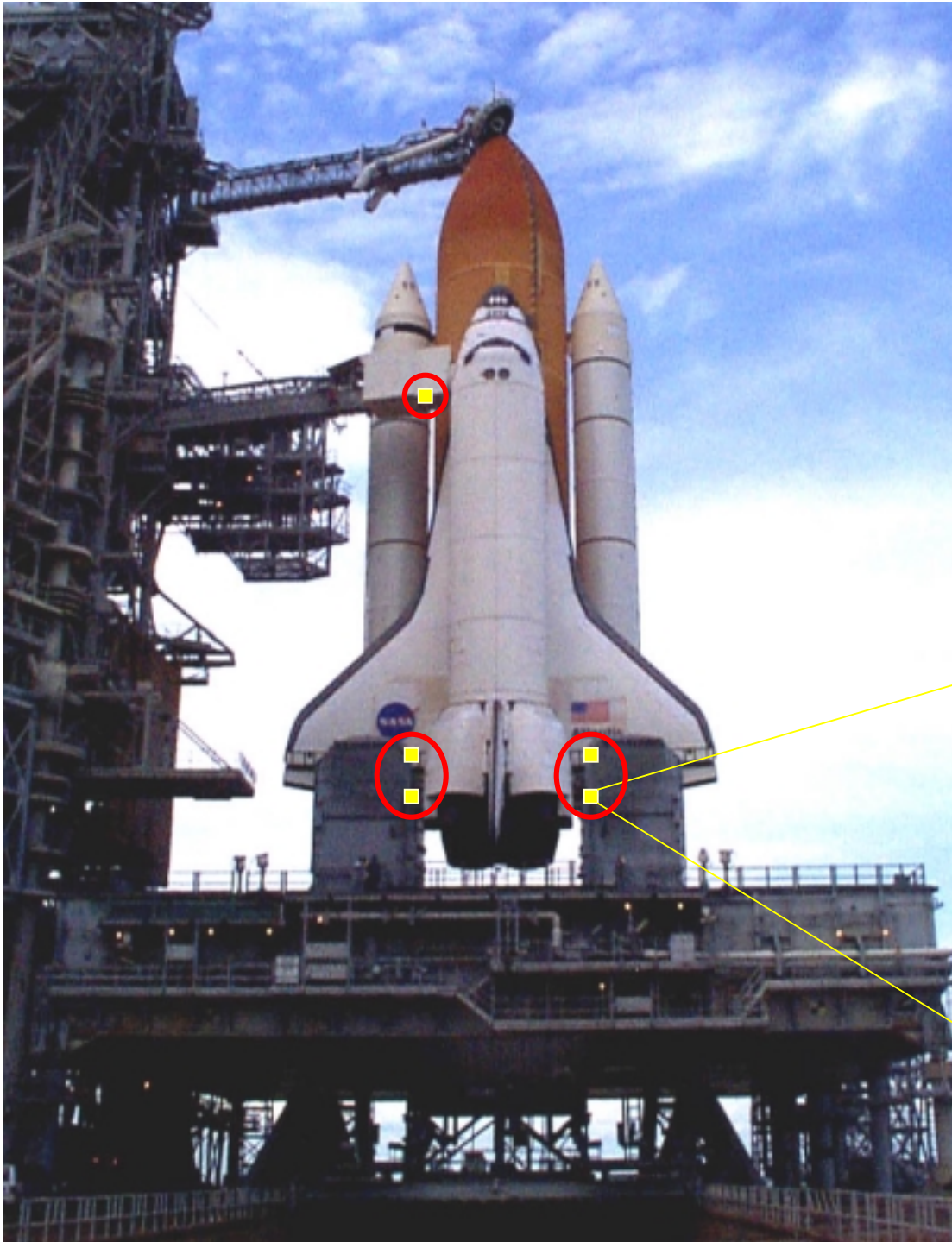


Current Leak Detection



Future

Leak Detection



QITMS

- 2 – 50 Da
- Fast Analysis
- Miniature & Rugged

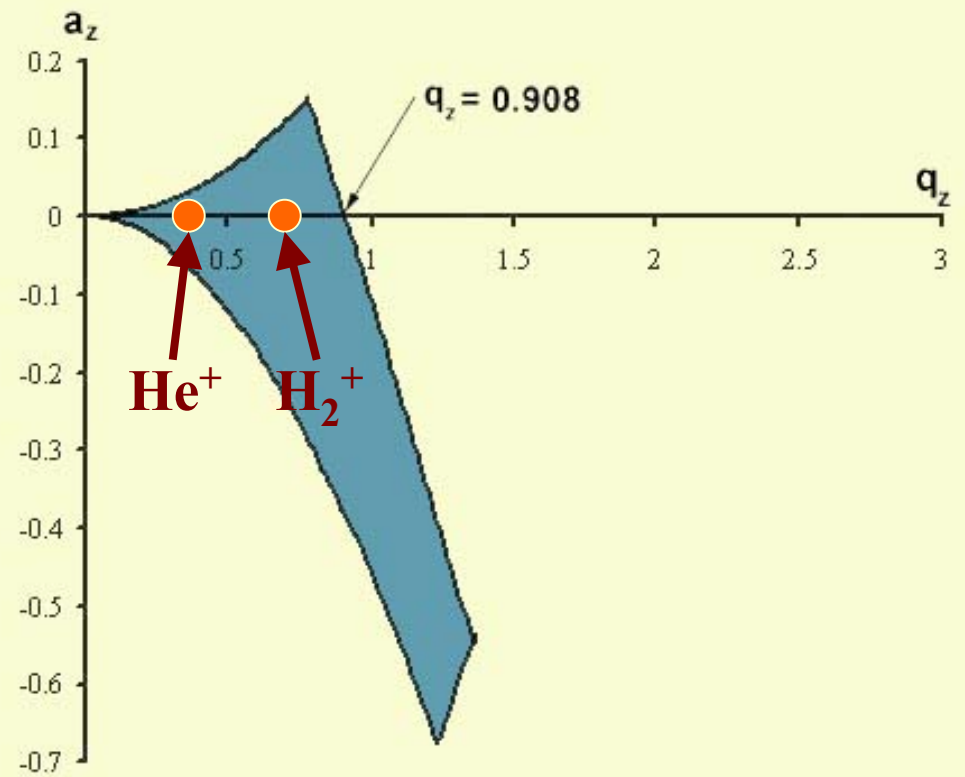


Modify Mass Range

$$m/z \, q_z = \frac{8eV_{RF}}{(r_0^2 + 2z_0^2)\Omega^2}$$

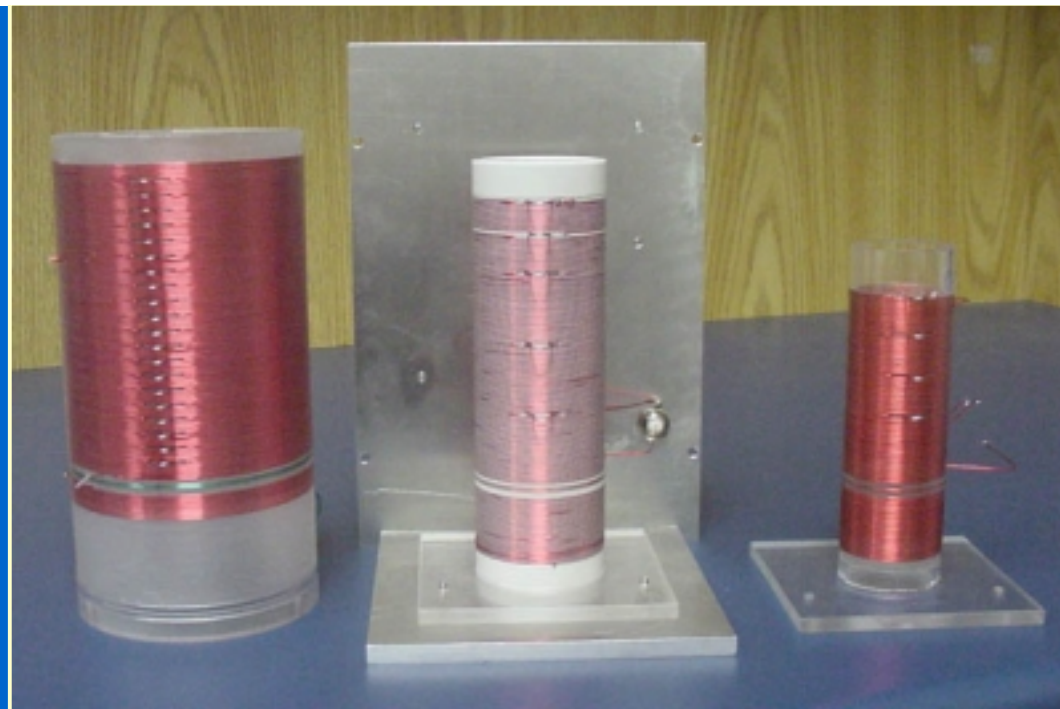
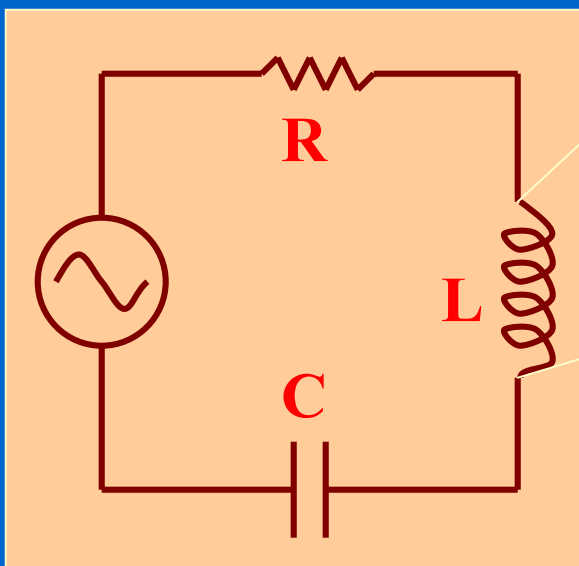
**Increase
Dimensions**

**Increase
Frequency**



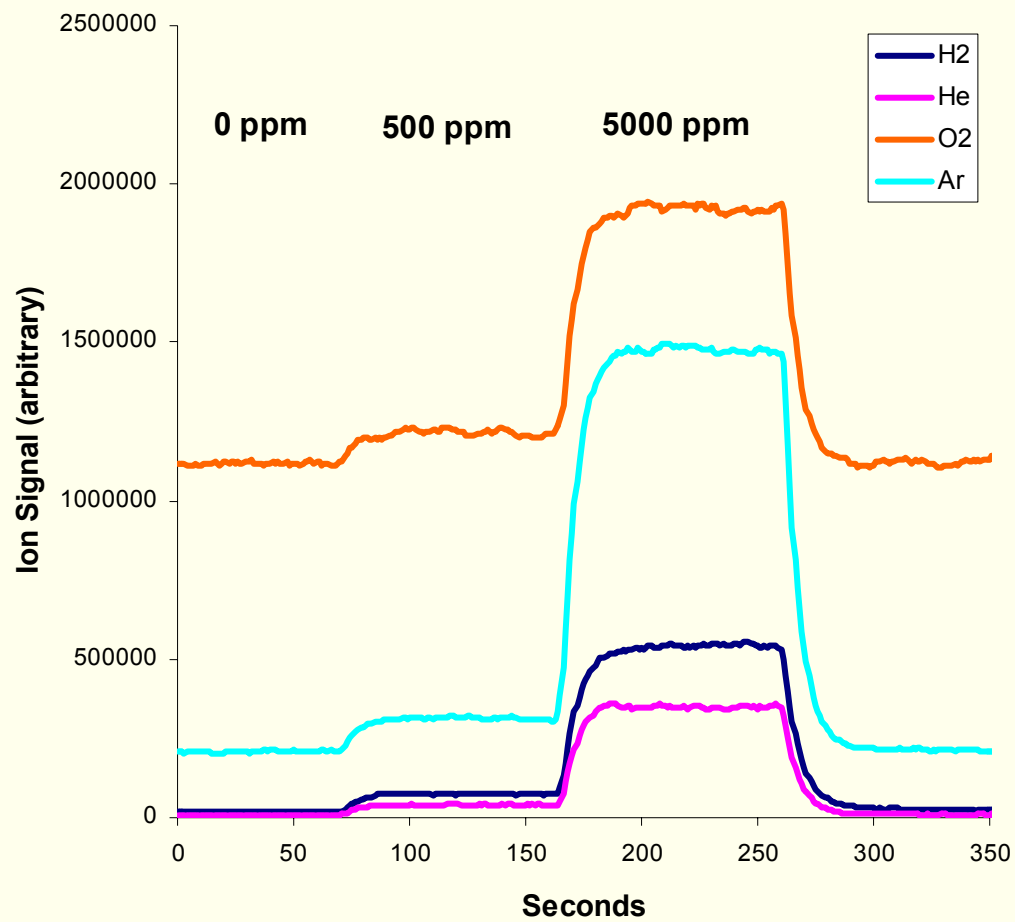
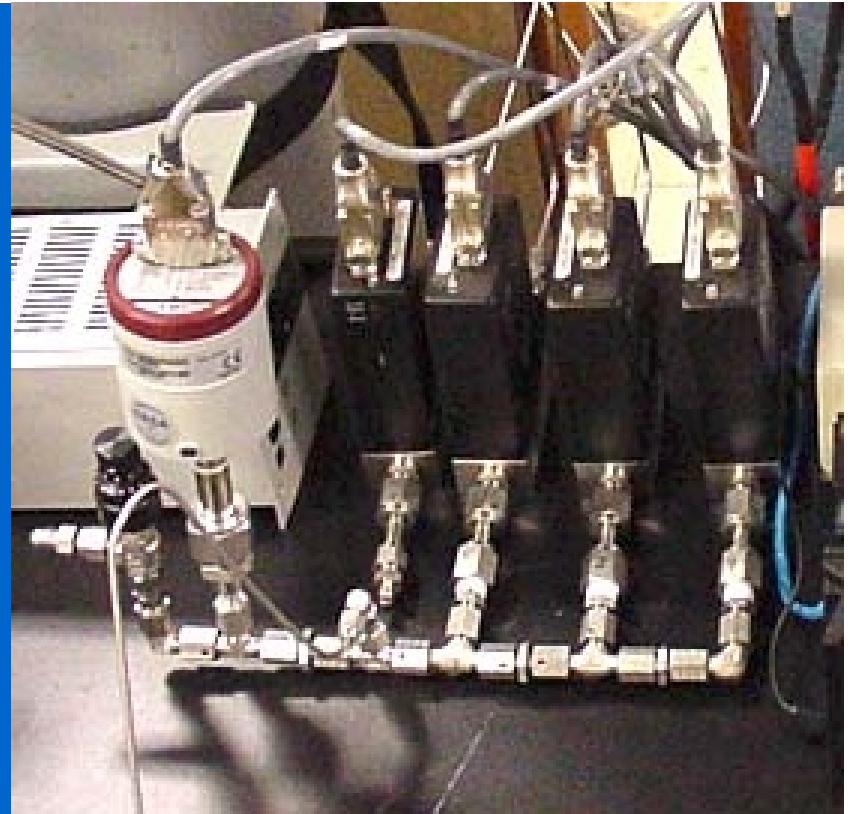
RF Circuit

RLC Network



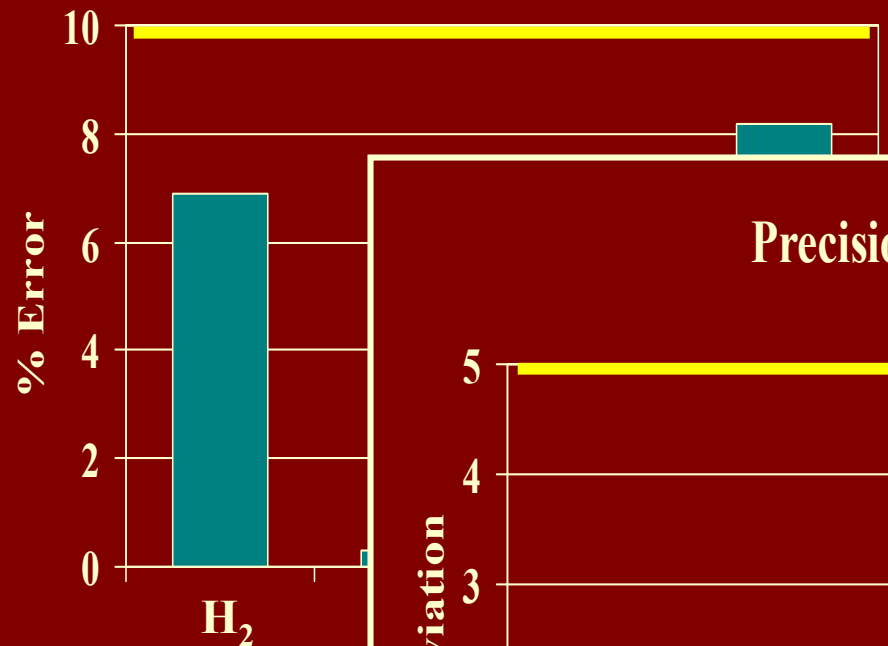
2.5 MHz $\rightarrow$ 2 to 60 Da

Testing

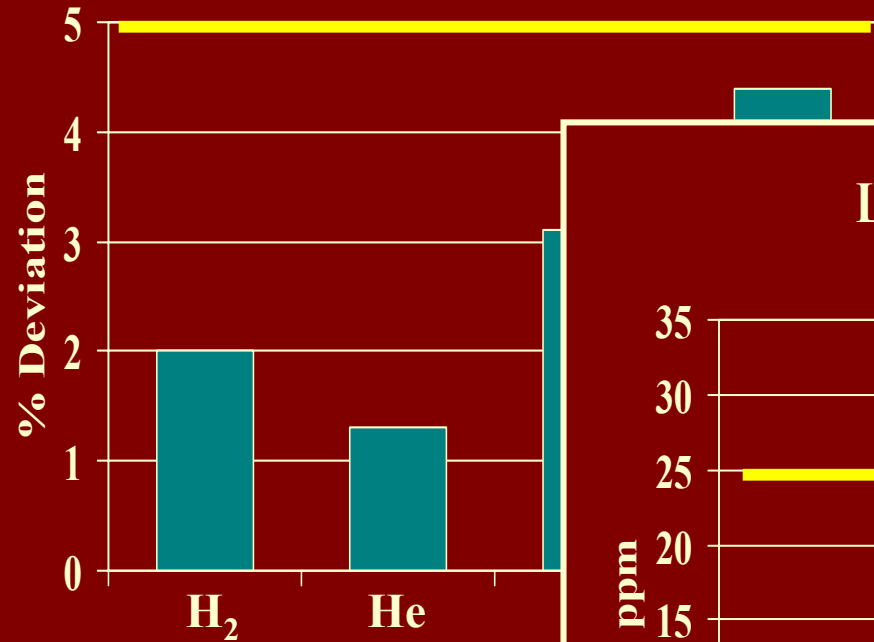


Quantitative Results

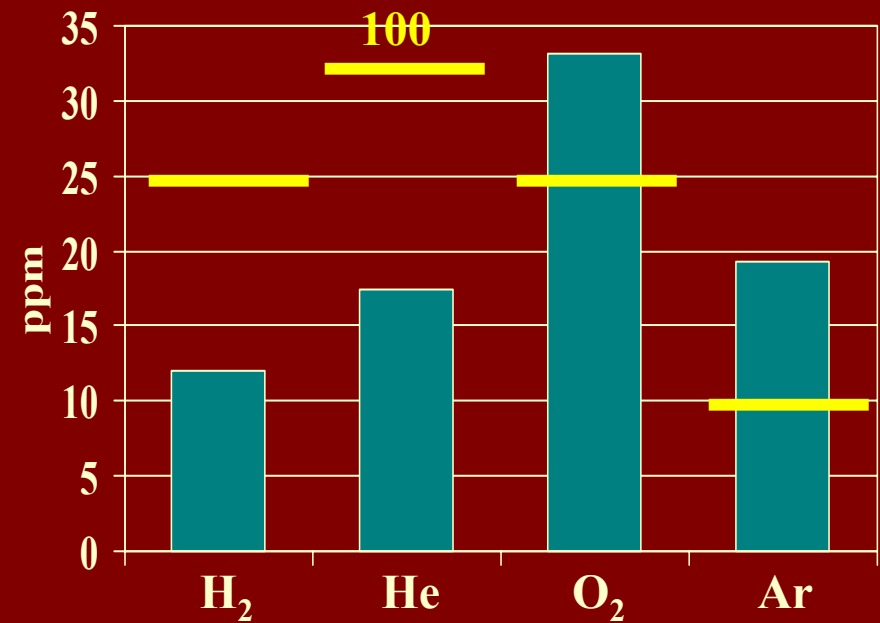
Accuracy



Precision



Limit of Detection



Trapped Ions

More Ions = More Signal

Or Not?

```
graph TD; A[Or Not?] --> B[Space Charge]; A --> C[Ion-Molecule Reactions];
```

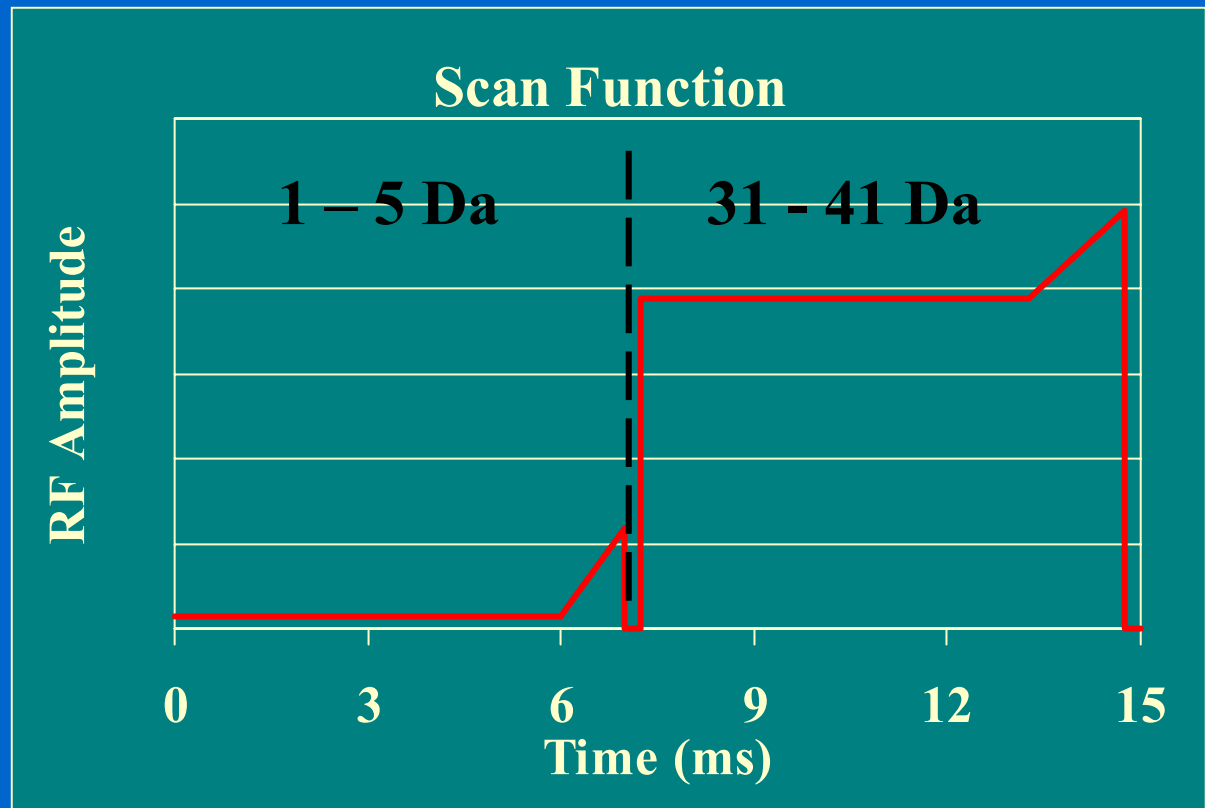
**Space
Charge**

**Ion-Molecule
Reactions**

Go Fast...

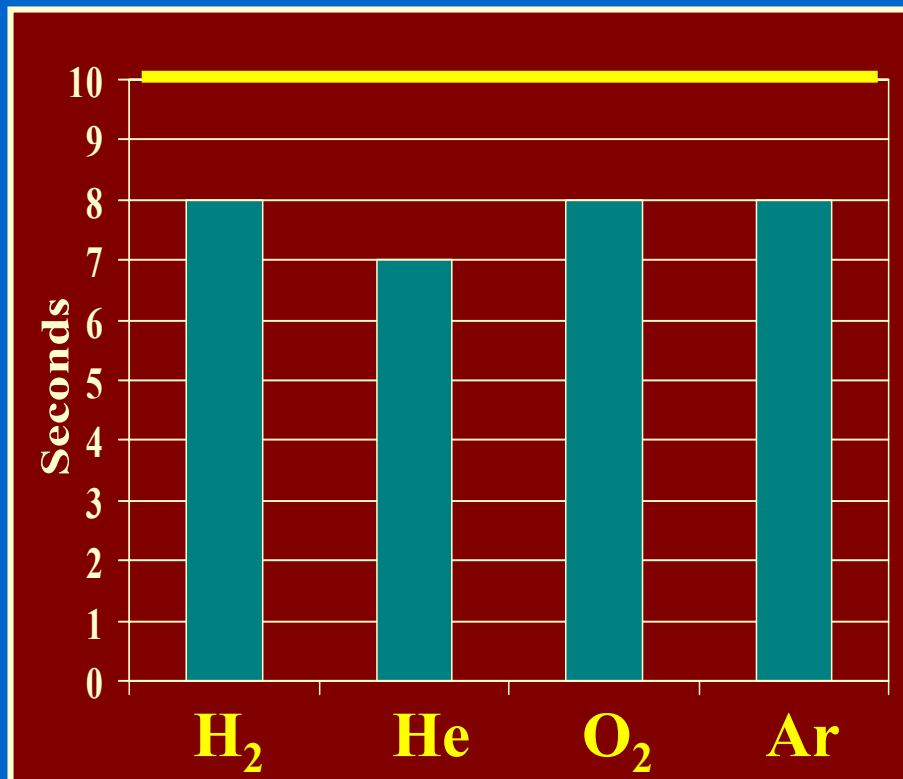
**Maximize
Signal**

**Minimize
Noise**

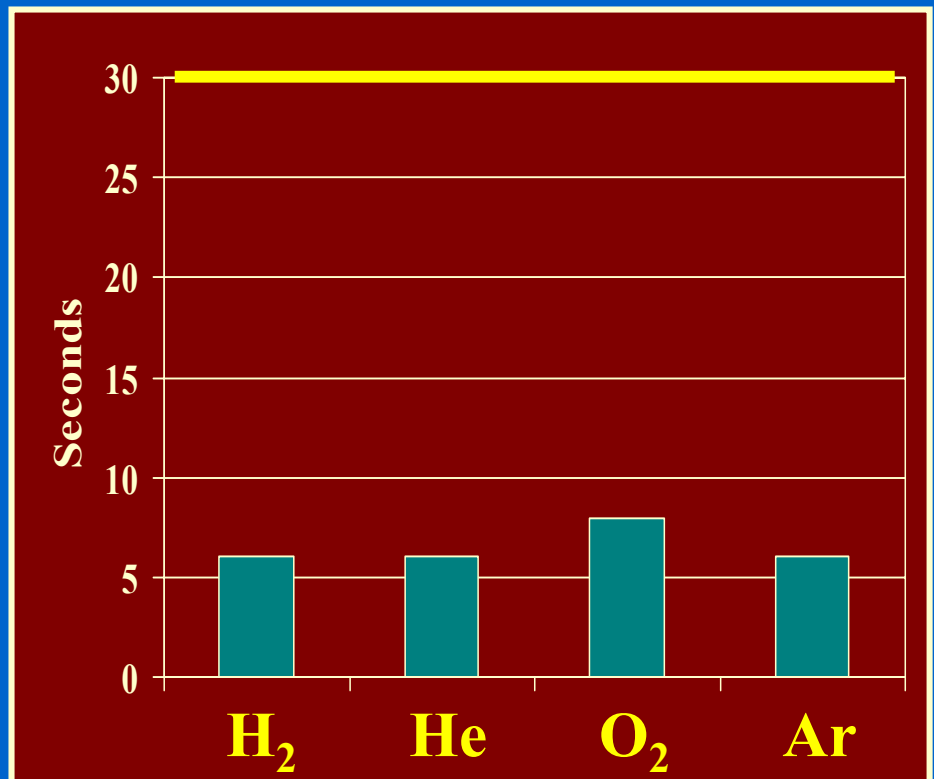


Quick Response

Response Time



Recovery Time



Miniature ?

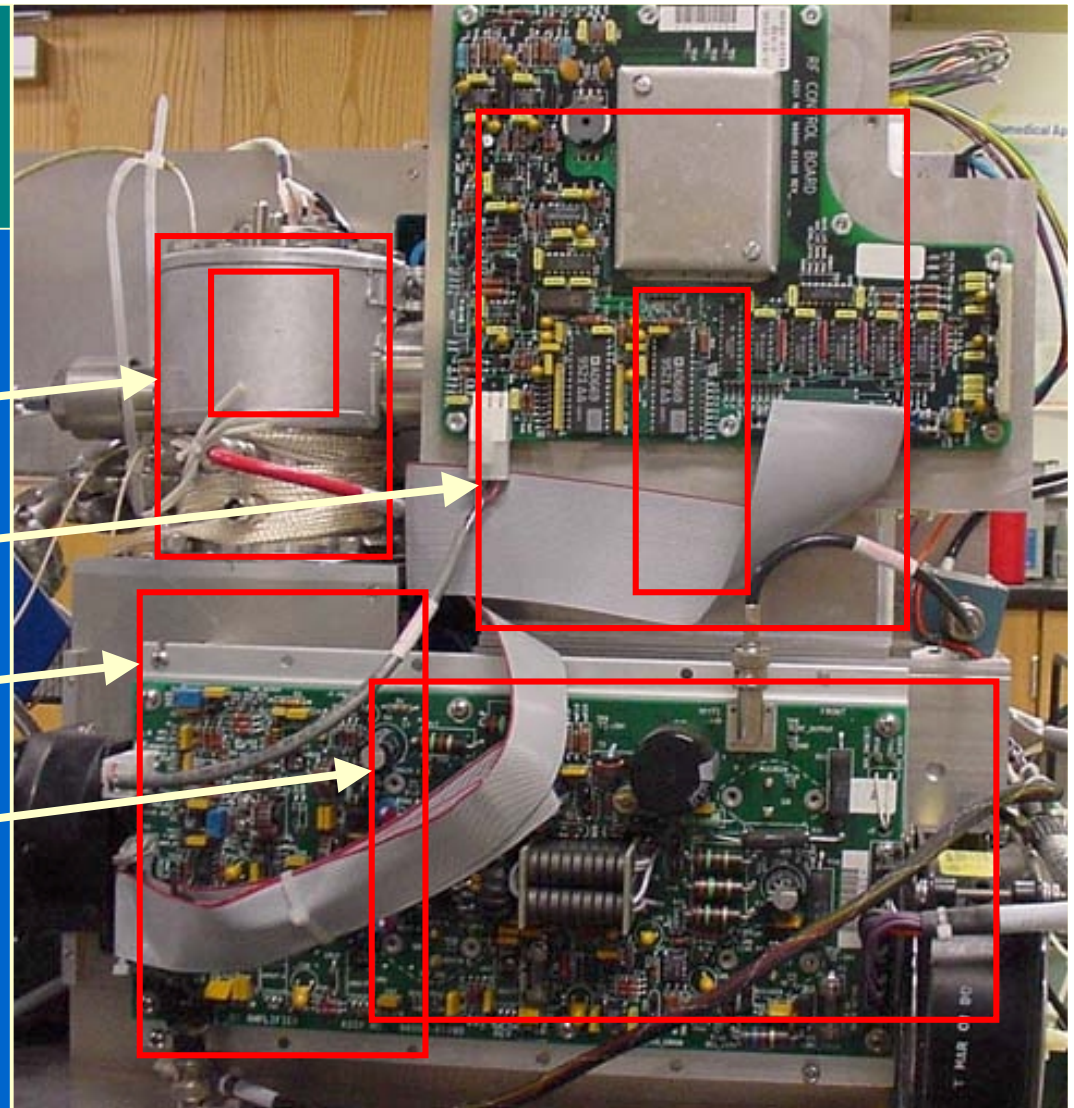
Volume

2 %

6.5 %

5 %

8 %



20 %

20 %

**Other
Electronics**

1/2 Size Trap

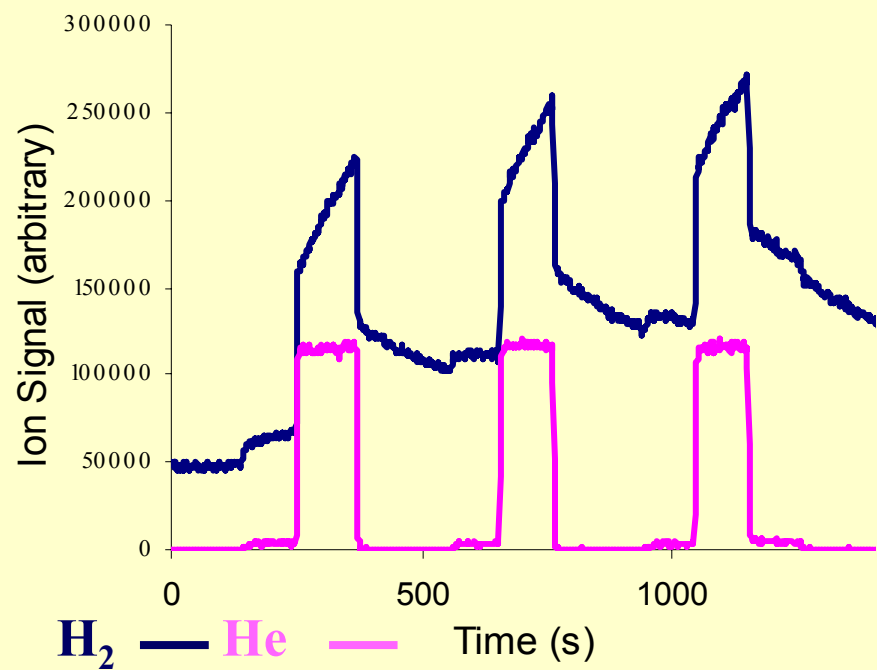
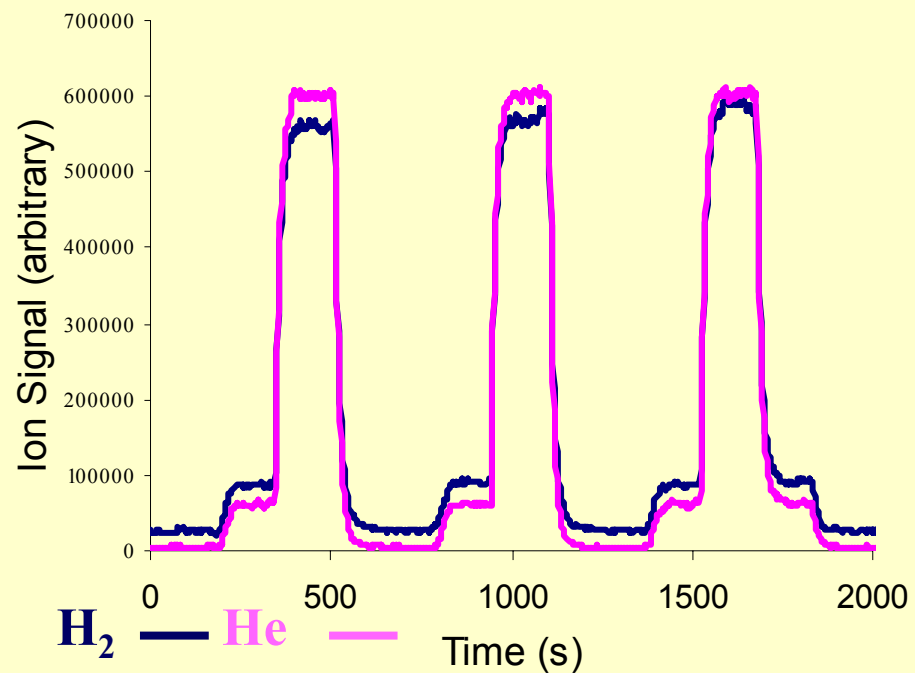
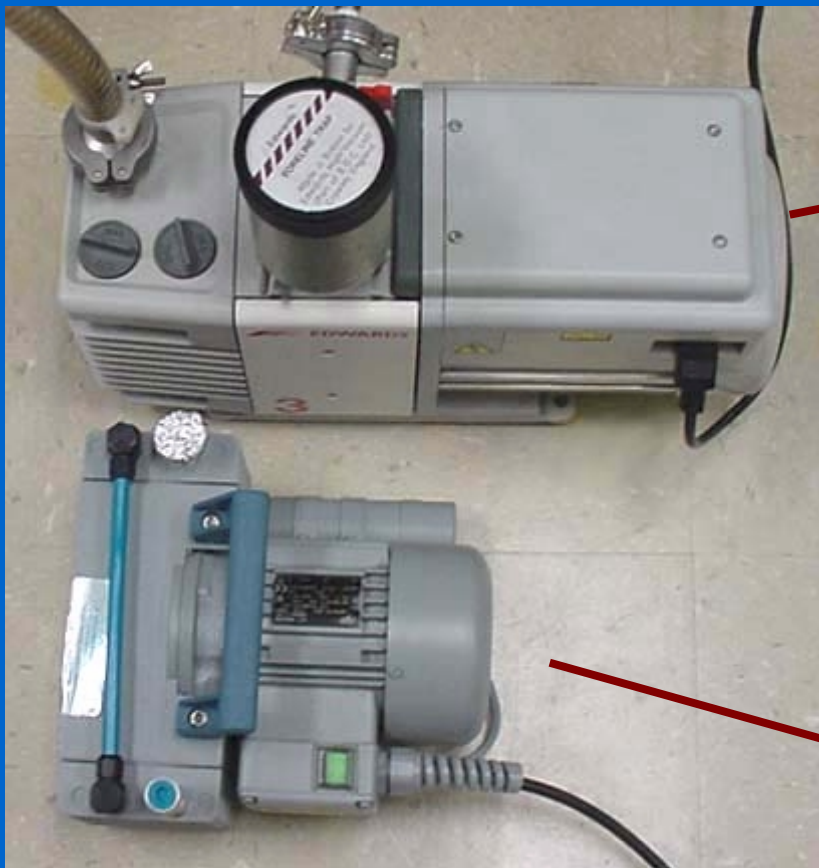
Reduce r_0 & z_0 by $1/2$:

- Trap volume reduced by $7/8$
- Less Pumping
- Coil volume reduced by $1/2$
- Less Power



**Will the
Ions Fit?**

Pumps



Rugged ?

- Solid State
- Source Filament
- Vacuum Pumps



Conclusions

QITMS

- Quantitative
- Fast Analysis

- Miniature
- Rugged



Acknowledgements

The funding for this project was provided under the Engineering Development Contract at Kennedy Space Center, Florida.